Cubic Regularization with Momentum for Nonconvex Optimization

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The paper in one minute

Question: Can momentum make cubic-regularized Newton practical faster without breaking its second-order convergence guarantee?
Model/Method: CRm: a cubic model step, an adaptive momentum extrapolation, then a monotone choice between candidates.
Finding: The method retains the worst-case $O(\varepsilon^{-3/2})$ iteration bound for approximate second-order stationarity and gains local quadratic convergence under an error-bound condition; experiments show acceleration.
Why we care: Momentum can be combined with curvature-aware updates, but empirical speed gains are distinct from improved worst-case rate exponents.

1. Start with cubic-regularized Newton

At a current point x_k, form a local model of an objective f:

$$m_k(s) = \nabla f(x_k)^T s + \tfrac{1}{2} s^T \nabla^2 f(x_k) s + \tfrac{M}{6}\|s\|^3.$$

The quadratic part captures gradient and curvature; the positive cubic term restrains the step when the Hessian is indefinite. Standard cubic regularization (CR) chooses a global minimizer s_{k+1} and sets $y_{k+1} = x_k + s_{k+1}$. This can exploit negative curvature at saddles, but each cubic step may be computationally expensive. Wang et al. ask whether adding momentum *after* such a step improves progress while retaining guarantees.

Worked example: how the cubic term escapes a saddle

Let $f(x) = -x^2/2 + x^4/4$, so $f'(0) = 0$, $f''(0) = -1$. At $x_0 = 0$, ordinary GD has update zero and cannot move. The local cubic model with $M = 2$ is

$$m_0(s) = -\tfrac{1}{2}s^2 + \tfrac{1}{3}|s|^3, \quad m_0'(s) = -s + s|s|.$$

Its minima occur at $s = \pm 1$; choosing $s = 1$ takes us to $y_1 = 1$ where $f(1) = -1/4$, lower than $f(0) = 0$. The method uses *negative curvature*, even though the gradient vanishes. This is a toy illustration, not an experiment reported by the authors.

A useful distinction. CRm's saddle escape comes from its cubic *second-order model*. Momentum is intended to accelerate the resulting sequence; it is not by itself a generic guarantee against saddle points.

2. The three steps of CRm (Algorithm 1)

Step 1: Cubic model. Solve the cubic subproblem to obtain $y_{k+1} = x_k + s_{k+1}$. **Step 2: Adaptive momentum.** Let y_k be the previous cubic candidate and choose

$$\beta_{k+1} = \min\{\rho, \|\nabla f(y_{k+1})\|, \|y_{k+1} - x_k\|\}, \qquad 0 < \rho < 1,$$

$$v_{k+1} = y_{k+1} + \beta_{k+1}(y_{k+1} - y_k).$$

If the current point is still far from stationarity, a larger motion is allowed. Near a stationary point both gradient norm and step norm shrink, so the extrapolation becomes small. **Step 3: Monotone safeguard.** Accept whichever candidate has less *actual* objective value:

$$x_{k+1} = \operatorname*{argmin}_{z \in \{y_{k+1}, v_{k+1}\}} f(z).$$

The safeguard is essential: arbitrary momentum may increase a nonconvex objective and invalidate a descent argument.

A tiny numerical check of the safeguard

Suppose the cubic step proposes $y_{k+1} = 0.8$, the previous cubic candidate was $y_k = 0.5$, and the adaptive coefficient is $\beta_{k+1} = 0.2$. Then $v_{k+1} = 0.8 + 0.2(0.3) = 0.86$. On $f(x) = -x^2/2 + x^4/4$:

$$f(0.8) = -0.2176, \qquad f(0.86) \approx -0.2330.$$

The method would accept 0.86, which has a lower objective. If extrapolation had a higher loss, it would instead accept 0.8. The values illustrate Algorithm 1's choice, not its parameter-setting theorem.

3. What is actually proved

Theorem 1: Under Assumption 1 and suitable M, CRm finds an ε-second-order stationary point in $O(\varepsilon^{-3/2})$ iterations. The relevant conditions are $\|\nabla f(x)\| \le \varepsilon$ and smallest Hessian eigenvalue bounded below on the order of $-\sqrt{\varepsilon}$. **Theorem 2:** Under an additional local error bound $\operatorname{dist}(x, X) \le \kappa\|\nabla f(x)\|$ near the stationary-point set X, the iterates converge locally at a quadratic rate. **Theorem 3:** A carefully inexact cubic subproblem still attains the same $O(\varepsilon^{-3/2})$ outer iteration order.

4. Assumptions, limitations and contributions

Assumption 1 requires f bounded below, bounded sublevel sets, and Lipschitz gradient/Hessian. The local quadratic theorem additionally needs the error-bound property, which is weaker than global strong convexity but does not hold for every nonconvex objective. The inexact version imposes explicit solver-accuracy conditions.

- **Contribution:** A carefully safeguarded momentum variant of cubic regularization with its global-order convergence guarantee preserved.
- **Empirical finding:** The paper's experiments report substantially faster decrease in objective value than conventional cubic regularization on tested problems.
- **Not proved:** A strictly better worst-case ε exponent; global minima in arbitrary neural networks; negligible cost per iteration. Cubic model minimization can remain expensive.

Survey connections

Carmon–Duchi (Ref 71) shows GD can solve cubic subproblems to useful accuracy: one route to making each CR step cheaper. **Dauphin (2014)** also uses Hessian curvature to address saddles, but with a different update. **Galli/Kalra (2025–26)** treat learning-rate and sharpness dynamics, not cubic Newton's local model. This paper fits Section V as a curvature-based intervention with a momentum extension.

Quick reference

Term	Meaning
$M\|\|s\|\|^3/6$	Stabilizes an indefinite quadratic Taylor model.
β_{k+1}	Adaptive momentum, limited by the gradient norm and cubic step length.
Monotone choice	Compare $f(y_{k+1})$ and $f(v_{k+1})$ before accepting.
$O(\varepsilon^{-3/2})$	Worst-case *outer iteration* order, not total wall-clock runtime.
Local error bound	Converts control of gradient norm into distance from the solution set.

Papers Master note

Wang et al. extend cubic-regularized Newton with adaptive momentum and a monotone safeguard. The method retains $O(\varepsilon^{-3/2})$ second-order-stationarity iteration complexity, gains local quadratic convergence under an error bound, and improves empirical optimization speed; cubic steps remain expensive.

Source: original UAI 2020 paper, Algorithm 1; Theorems 1–3.